

SECURITY INCIDENT REPORT

SSH Brute Force Attack — LINUX01

Incident ID: IR-2026-001 | Classification: PORTFOLIO SAMPLE | Status: CLOSED

Severity	Incident Date	Analyst	Detection Time	Organization	Resolution Time	Affected System	Total Duration
HIGH	June 1, 2026	Ramon Vera	21:11 MDT	Personal SOC Lab	21:30 MDT	LINUX01 (192.168.100.30)	~19 minutes

1. Executive Summary

On June 1, 2026 at approximately 21:11 MDT, Splunk SIEM detected an automated SSH brute force attack originating from internal host KALI01 (192.168.100.128) targeting LINUX01 (192.168.100.30) on port 22. A total of 2,334 failed authentication attempts were recorded over approximately 19 minutes. No successful authentication was achieved. The attack was detected by the **Brute Force SSH Detection** alert configured in Splunk. The source IP has been identified and the incident is considered closed with no data breach or system compromise.

2. Incident Timeline

Time (MDT)	Event	Source
21:08	Hydra brute force tool initiated from KALI01	auth.log
21:08:01	First failed SSH attempt recorded — invalid user	auth.log
21:08 – 21:27	2,334 failed authentication attempts generated	auth.log
21:27	Brute Force SSH Detection alert triggered in Splunk	Splunk Alert
21:27	Analyst investigation initiated	Manual
21:30	Source IP identified as KALI01 (192.168.100.128)	Splunk Search
21:30	No successful logins confirmed — incident closed	auth.log

3. Affected Systems

System	IP Address	OS	Role	Status
LINUX01	192.168.100.30	Ubuntu 24.04 LTS	Target — SSH Server	Not Compromised
KALI01	192.168.100.128	Kali Linux	Attack Source	Identified

4. Evidence & Investigation

4.1 Splunk Detection

The following Splunk search confirmed the attack details:

```
source="/var/log/auth.log" | rex "from (?P<src_ip>\d+\.\d+\.\d+\.\d+)" | stats count(eval(match(_raw,"Failed"))) as failures count(eval(match(_raw,"Accepted"))) as successes by src_ip | where failures > 3
```

Field	Value	Significance
Source IP	192.168.100.128 (KALI01)	Internal host used as attack source
Target	192.168.100.30 port 22 (LINUX01)	SSH service targeted
Failed Attempts	2,334	Consistent with automated brute force tool
Successful Logins	0	No credentials compromised
Attack Duration	~19 minutes	Automated — too fast for manual attempt
Tool Used	Hydra (inferred from pattern)	Rapid sequential attempts from single source

4.2 Sample Log Entries from auth.log

The following entries represent the attack pattern observed in /var/log/auth.log:

```
2026-06-01T21:08:01 LINUX01 sshd[85123]: Failed password for invalid user wronguser from 192.168.100.128 port 54231 ssh2
2026-06-01T21:08:02 LINUX01 sshd[85124]: Failed password for invalid user wronguser from 192.168.100.128 port 54232 ssh2
2026-06-01T21:08:03 LINUX01 sshd[85125]: Failed password for invalid user wronguser from 192.168.100.128 port 54233 ssh2
[ 2,331 additional failed attempts omitted ]
```

5. Impact Assessment

Category	Impact	Notes
Data Breach	None	No successful authentication occurred
System Integrity	None	LINUX01 operating normally
Service Availability	Minimal	SSH service remained available throughout
Credential Exposure	None	No passwords compromised
Network Impact	Low	Attack traffic limited to internal lab network

6. Recommendations

Priority	Recommendation	Rationale
High	Install fail2ban on LINUX01	Automatically block IPs after repeated failed SSH attempts — prevents brute force
High	Disable SSH password authentication — require key-based auth only	Eliminates brute force as an attack vector entirely — keys cannot be guessed
Medium	Change SSH port from 22 to non-standard port	Reduces automated scan and brute force noise significantly
Medium	Implement Linux account lockout via PAM faillock	Mirrors Windows AD lockout policy for Linux local accounts
Low	Review and audit all accounts with SSH access	Ensure only authorized users have SSH access to LINUX01

7. Lessons Learned

This incident demonstrates the importance of SIEM monitoring in detecting automated attacks. Without Splunk ingesting auth.log in real time, this brute force attack would have gone undetected. The Brute Force SSH Detection alert fired correctly and the investigation confirmed no breach occurred.

Key takeaway: Linux systems do not have account lockout protection by default. Unlike Windows Active Directory which enforces lockout after 5 failed attempts, Linux SSH will accept unlimited authentication attempts unless explicitly configured otherwise via fail2ban or PAM faillock. This gap should be addressed as a priority hardening action.

Analyst	Ramon Vera	Title	Network & Systems Administrator
Date	June 1, 2026	Incident Status	CLOSED — No Breach Confirmed